Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-7

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following stands for LP in LPWAN?

    a. Large Power
    b. Light Power
    c. Low Power
    d. Long Range

Answer: (c)

Justification: Please refer to Slide 3 of Lecture 1 of Week 7

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2. “We need to pay a license fee to use LoRa in the 868 MHz radio frequency band in Europe.” State True or False.

    a. True
    b. False

Answer: (b)

Justification: Please refer to Slide 4 of Lecture 1 of Week 7

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3. Which of the following is true about the SIGFOX network in terms of messages and payload?

    a. Up to 50 messages per day with a payload of 8 bytes per message
    b. Up to 75 messages per day with a payload of 10 bytes per message
    c. Up to 140 messages per day with a payload of 12 bytes per message
    d. Up to 200 messages per day with a payload of 15 bytes per message

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 1 of Week 7

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4. Which library (header file) is used for interfacing the DHT22 sensor, as demonstrated in the Hands On Industrial Environment Monitoring project?

a. DHT.h
b. DHT22.h
c. DHT11.h
d. Adafruit_DHT.h

Answer: (a)

Justification: Please refer to Slide 18 of Lecture 1 of Week 7

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5. Which of the following topologies are supported by Zigbee?

a. Point-to-Point and Star
b. Bus and Ring
c. Tree, and Mesh
d. Ring and Mesh

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 2 of Week 7

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6. The physical and MAC layers of the Zigbee communication protocol is based on _____. Fill in the blanks.

a. IEEE 802.11
b. IEEE 802.15.4
c. IEEE 802.3
d. IEEE 802.16

Answer: (b)

Justification: Please refer to Slide 5 of Lecture 2 of Week 7

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7. “Xbee is a mesh communication protocol based on the IEEE 802.15.4 whereas Zigbee is the product that uses the Xbee communication protocol for radio communication.” State true or false.

a. True
b. False

Answer: (b)

Justification: Please refer to Slide 7 of Lecture 2 of Week 7

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8. Which of the following functions is performed by the coordinator node in a Zigbee network?
    a. Data relay
    b. End device connector
    c. Acts as the root of the network and bridge between different networks
    d. Control plane

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 2 of Week 7

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9. Which of the following is/are the data characteristics of IIoT processing?

a. Polymorphism
b. Temporal relationships in data
c. Correlation in space, time, and other dimensions
d. All of the above

Answer: (d)

Justification: Please refer to Slide 3 of Lecture 3 of Week 7

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10. Which of the following is false about the rule-based engine?

a. Extract causal and temporal patterns using predefined rules
b. Handles multiple data streams and correlates them to provide meaningful output
c. Cannot process data in near real-time
d. Performs decision-making processes based on predefined rules

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 3 of Week 7

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11. Which of the following represents the limitation of current CEP systems?

a. Manual threshold specification
b. Run-time update of threshold not possible
c. Not context-aware
d. All of the above

Answer: (d)

Justification: Please refer to Slide 13 of Lecture 3 of Week 7

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12. Which of the following options is false about the services rendered by the Smart Water Management System?

a. Entirely non-replicable services
b. Fully customizable services
c. Application-specific services
d. Entirely replicable services

Answer: (a)

Justification: Please refer to Slide 4 of Lecture 4 of Week 7

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13. “iRobot-Factory is a cognitive manufacturing factory which applies cognitive intelligence and edge computing for improved manufacturing.” State true or false.

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 14 of Lecture 4 of Week 7

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14. Which of the following is false about the design of the SmartSantander?

a. Heterogeneity
b. Reliability
c. Mobility
d. Unrealistic experimentation

Answer: (d)

Justification: Please refer to Slide 11 of Lecture 4 of Week 7

15. Which of the following represents a type of Industrial Control System?

a. Programmable Logic Compiler
b. Decentralized Control Suite
c. Supervisory Control and Data Acquisition
d. Distributed Compiled System

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 5 of Week 7